

HIGHER REGULARITY FOR RELAXED MICROMORPHIC MODELS: FROM SMOOTH TO LIPSCHITZ DOMAINS

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The relaxed micromorphic model is a generalized continuum model designed to describe the effective behaviour of microstructured materials and metamaterials. It couples the classical displacement field with a non-symmetric microdistortion tensor P and, from the analytical point of view, combines an elasticity-type system with a Maxwell-type system.

A characteristic feature of the model is that the curvature energy depends on $\text{Curl } P$ rather than on the full gradient ∇P . Consequently, weak solutions are naturally sought in the space

$$H(\text{Curl}; \Omega) = \{P \in L^2(\Omega) : \text{Curl } P \in L^2(\Omega)\},$$

while the energy does not directly control all weak derivatives of P . This naturally raises the question of whether one can recover control of the full gradient ∇P from the structure of the system.

I will first briefly introduce the relaxed micromorphic model and recall earlier existence and higher-regularity results, including local regularity and global H^1 -regularity of the microdistortion on smooth domains. The main part of the talk will be devoted to a global regularity result for a physically nonlinear version of the model on bounded Lipschitz domains. Since standard difference-quotient arguments are not suitable in this setting, the proof is based on modified finite differences constructed from inner variations together with a Piola-type transformation adapted to the Curl structure. This yields, for every $\varepsilon > 0$,

$$u \in H^{\frac{3}{2}-\varepsilon}(\Omega), \quad P, \text{Curl } P \in H^{\frac{1}{2}-\varepsilon}(\Omega).$$

The results presented in the talk are based on joint work with Dorothee Knees and Patrizio Neff.

[1] S. Owczarek, I.-D. Ghiba, P. Neff, A note on local higher regularity in the dynamic linear relaxed micromorphic model, *Mathematical Methods in the Applied Sciences*, 2021, 44(18), 13855–13865.

[2] D. Knees, S. Owczarek, P. Neff, A local regularity result for the relaxed micromorphic model based on inner variations, *Journal of Mathematical Analysis and Applications*, 2023, 519(2), 126806.

[3] D. Knees, S. Owczarek, P. Neff, A global higher regularity result for the static relaxed micromorphic model on smooth domains, *Proceedings of the Royal Society of Edinburgh: Section A Mathematics*, 2026, 156(2), 436–450.

[4] D. Knees, S. Owczarek, P. Neff, Global regularity in a nonlinear relaxed micromorphic continuum on Lipschitz domains, *Calculus of Variations and Partial Differential Equations*, 2025, 64, 1–22.